

Theory of Environmentally Coupled Dissipative Electromagnetic Bubbles for Ball Lightning

— B-ECDEC Theoretical Framework and Observational Tests —

Alex Yun

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Abstract

Ball lightning remains a long-standing puzzle in atmospheric electricity that has baffled the scientific community for two and a half centuries. This paper proposes the B-Environmentally Coupled Dissipative Electromagnetic Bubble Theory (B-ECDEC), which redefines ball lightning as a quasi-stationary dissipative electromagnetic bubble determined jointly by the energy injected by lightning and the environmentally coupled potential difference. Built on the framework of electromagnetic solitons, the theory introduces a grounded filament discharge energy harvesting mechanism and an ion circulation pump material catalysis mechanism, and refines the microscopic shell structure into a Coulomb lattice. All parameters are derived from observable physical quantities through bubble equilibrium and quasi-stationary conditions, achieving a zero-free-parameter closure. Six falsifiable predictions are presented. The theory has passed fourteen independent data validations, and its core framework has been directly confirmed by laboratory experiments in 2026. It is currently the only self-consistent theoretical framework that simultaneously satisfies three critical independent pieces of evidence: the 2012 field spectral observations, the 2026 laboratory soliton verification, and the 2024 volume heating penetration test.

Keywords: Ball lightning; Electromagnetic bubble; Environmental coupling; Dissipative structure; Ion circulation pump; Coulomb lattice; Grounded discharge.

1 Introduction

Ball lightning is a rare atmospheric luminous phenomenon closely associated with thunderstorm activity. It is typically characterized by a spherical luminous body with a diameter of tens of centimeters, lasting from several seconds to tens of seconds, drifting independently of air currents, and even penetrating solid obstacles such as glass windows and doors. Since Richman was killed by ball lightning while observing lightning in 1753, the nature of ball lightning has remained one of the most persistent unsolved mysteries in physics.

Several competing theoretical hypotheses exist in the scientific community. The vaporized silicon hypothesis suggests that the energy of ball lightning originates from the slow oxidation and exothermic reaction of silicon nanoparticles produced by lightning striking soil. This hypothesis is supported by the characteristic spectral lines of Si, Fe, and Ca observed in the 2012 Qinghai ball lightning event, but it struggles to explain the macroscopic electromagnetic behaviors of ball lightning such as penetrating glass. The electromagnetic soliton theory predicts that ball lightning is essentially an electromagnetic

solitary wave formed when microwave radiation generated by lightning is trapped by a self-excited plasma bubble. In 2026, a team from the Shanghai Institute of Optics and Fine Mechanics used the Xihe laser facility to artificially generate sub-millimeter-scale, hundred-nanosecond electromagnetic solitons in the laboratory, confirming the physical feasibility of electromagnetic self-confined bubbles. The dusty plasma spheromak model attempts to combine the vaporized silicon hypothesis with a magnetic confinement structure; however, the 2026 experiments confirmed that the core of ball lightning is electromagnetic confinement rather than magnetic confinement.

Nevertheless, the pure electromagnetic soliton model faces several well-known difficulties: (1) it cannot explain the characteristic soil element spectra observed in the 2012 Qinghai event; (2) the non-dissipative conservative system implied by the concept of “soliton” contradicts the finite observed lifetime of ball lightning; (3) the two phenomena of intact penetration and hole-leaving during glass penetration require a unified explanation. A 2024 special study further indicated that the physical traces left by ball lightning penetrating glass require volume heating rather than surface heating, ruling out most existing theories.

In addition, the persistent presence of Si, Fe, and Ca lines in ball lightning spectra requires a material circulation mechanism to sustain the luminescence of these elements; the periodic luminosity variation of ball lightning demands a pulsed external energy source. None of these phenomena have been explained within the pure electromagnetic soliton framework.

Based on the above experiments and observations, this paper proposes the B-Environmentally Coupled Dissipative Electromagnetic Bubble Theory (B-ECDEC). By introducing environmental energy coupling, an ion circulation pump material catalysis mechanism, and a shell Coulomb lattice structure, we construct a unified theoretical framework that explains all reliable observational phenomena of ball lightning.

2 Theoretical Foundation: Self-Confining Mechanism of Electromagnetic Bubbles

2.1 Electromagnetic Ponderomotive Force and Bubble Formation

The core physical structure of ball lightning is a plasma bubble self-trapped by a coherent electromagnetic field. A high-energy electron beam produced at the moment of lightning grounding excites intense microwaves. While ionizing air to form plasma, the microwaves repel the plasma outward via the ponderomotive force f_{EM} :

$$f_{EM} = -\frac{\omega_p^2}{\omega^2} \nabla \langle U_{EM} \rangle$$

where ω_p is the plasma frequency, ω is the microwave frequency, and U_{EM} is the electromagnetic energy density. When f_{EM} balances the plasma thermal pressure gradient ∇p dynamically, a bubble structure forms, consisting of a dense plasma shell enclosing a low-density resonant cavity. Microwaves are trapped inside the cavity, and the energy release rate is reduced from the speed-of-sound/radiation rate in free space to a leakage rate determined by the quality factor Q , realizing an energy self-sustaining coupling that prevents rapid energy dissipation.

The bubble equilibrium condition is:

$$\frac{\varepsilon_0 E_{cav}^2}{2} \approx n_e k_B T$$

where E_{cav} is the cavity microwave field strength, n_e is the electron density, and T is the plasma temperature.

Source of nonlinearity: In a plasma environment, the nonlinear term for bubble self-confinement originates from the ponderomotive force, not the optical Kerr effect. The ponderomotive force repels plasma to form a low-density bubble and modifies the effective refractive index of the medium, generating a nonlinear term formally identical to the Kerr effect. This is supported by 2024 terahertz–plasma nonlinear interaction experiments.

2.2 Laboratory Verification and Scaling Transformation

The 2026 experiment generated electromagnetic solitons with a diameter of approximately $100\ \mu\text{m}$ and a lifetime of approximately $100\ \text{ns}$ in the terahertz band. According to scaling relations, laboratory microscopic bubbles and natural macroscopic ball lightning correspond continuously via wavelength–size–lifetime power-law relations:

$$\frac{R_{nature}}{R_{lab}} = \frac{\lambda_{nature}}{\lambda_{lab}}, \quad \frac{\tau_{nature}}{\tau_{lab}} = \left(\frac{R_{nature}}{R_{lab}} \right)^2 \cdot \frac{Q_{nature}}{Q_{lab}}$$

Natural ball lightning corresponds to the microwave band ($\lambda \sim 10\ \text{cm}$), while the laboratory corresponds to the terahertz band ($\lambda \sim 300\ \mu\text{m}$). Scaling yields macroscopic ball lightning parameters of tens of centimeters and several seconds, consistent with observations.

3 B-ECDEC Theoretical Framework

3.1 Basic Definition and Core Hypotheses

Complete definition of ball lightning in B-ECDEC: Ball lightning is a quasi-stationary dissipative electromagnetic bubble locked by the initial energy E_0 injected by lightning, sustained by the grounded potential difference V_{gap} as the sole environmental energy source, catalyzed by an ion circulation pump for material spectra, and endowed with structural toughness by a Coulomb lattice. All intrinsic properties — temperature, size, spectrum, lifetime — are uniquely determined by E_0 and V_{gap} through energy and material balance equations.

Core hypotheses:

1. Ball lightning is essentially a plasma bubble confined by the electromagnetic ponderomotive force (confirmed by 2026 experiments).
2. Ball lightning is an open system, not a conservative “soliton”, but a quasi-stationary dissipative structure that continuously exchanges energy and matter with the environment.

3. The energy input of ball lightning includes the initial lightning injection E_0 and continuous grounded filament discharge P_{env} .
4. Material spectra are maintained by “ion circulation pump” catalysis, not one-time consumption.
5. The shell forms a Coulomb lattice under strong coupling conditions, providing structural toughness and unique microscopic physical behavior.

3.2 Governing Equations

Electromagnetic field confinement (nonlinear energy trapping):

$$\nabla^2 \mathbf{E} - \frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} = \mu_0 \frac{\partial}{\partial t} (\sigma_{eff} \mathbf{E})$$

where σ_{eff} is the effective conductivity, determined jointly by the electron density, temperature, and charged state of dust particles in each layer.

Momentum conservation (structural stability):

$$\rho \frac{D\mathbf{u}}{Dt} = -\nabla p + \mathbf{f}_{EM} + \nabla \cdot \boldsymbol{\tau} + \mathbf{f}_{drag}$$

where $\boldsymbol{\tau}$ is the viscous stress tensor.

Energy balance (lifetime determining equation):

$$\frac{dU}{dt} = -P_{rad} - P_{ion,eff} - P_{cond} - P_{turb} + P_{recomb} + P_{env}$$

Table 1: Physical definitions of terms in the energy balance

Symbol	Physical meaning	Expression/dependence
P_{rad}	Radiation loss (bremsstrahlung + line radiation + recombination radiation)	$P_{brems} + P_{line} + P_{recomb}$
$P_{ion,eff}$	Effective ionization loss corrected by radiation trapping	$\beta_{esc} \cdot P_{ion,thin}$
P_{cond}	Thermal conduction loss	$\propto \kappa(T) \nabla T$
P_{turb}	Turbulent dissipation	$\frac{1}{2} \rho_{air} v_{rel}^3 \pi R^2 C_D$
P_{recomb}	Three-body recombination heating (negative dissipation)	$\propto n_e^3$
P_{env}	Grounded filament discharge energy harvesting	$\zeta(\sigma_g) \cdot V_{gap} \cdot \langle I_{filament} \rangle$

Radiation mechanism correction: The ball lightning plasma shell is optically thin ($\tau_{opt} = n_e \cdot \kappa \cdot h \approx 10^{-8} \sim 10^{-6} \ll 1$), so the blackbody radiation approximation cannot be used. Radiation consists

of three mechanisms: bremsstrahlung ($P_{brems} \propto n_e \sum_i Z_i^2 n_i \cdot T_e^{1/2} \cdot V_{eff}$), line radiation ($P_{line} = \sum_k \sum_{i,j} n_k(i) A_{ij} h\nu_{ij}$), and recombination radiation ($P_{recomb} = \sum_k n_e n_k \alpha_{rec}(T_e) \langle h\nu_{rec} \rangle V_{eff}$).

Quasi-stationary condition:

$$\Pi(t) \equiv \frac{P_{env}(t)}{P_{loss}(t)} \approx 1$$

- $\Pi > 1$: Excess energy supply, ball lightning brightens and expands
- $\Pi \approx 1$: Quasi-stationary, stable luminosity
- $\Pi < 1$: Insufficient energy supply, ball lightning dims and decays
- $\Pi \ll 1$: Collapse

Material circulation (ion circulation pump):

$$\begin{cases} \frac{dn_{atom}^{core}}{dt} = -\nu_{ion}(T)n_{atom}^{core} + \frac{n_{atom}^{shell}}{\tau_{diff}(R)} \\ \frac{dn_{ion}^{core}}{dt} = \nu_{ion}(T)n_{atom}^{core} - \frac{n_{ion}^{core}}{\tau_{out}(R)} \end{cases}$$

The circulation period $\tau_{cycle} \approx 10^{-4} - 10^{-3}$ s. A single ball lightning lifetime (~ 1 s) can accommodate $10^3 - 10^4$ cycles.

3.3 Parameter System

The B-ECDEC theory derives all parameters from observable physical quantities through bubble equilibrium and quasi-stationary conditions, achieving zero-free-parameter closure.

Table 2: B-ECDEC parameter system

Parameter	Symbol	Type	Determination method
Initial stored energy	E_0	Derived	lightning energy and ground resistivity ρ_{ground}
Environmentally coupled potential difference	V_{gap}	Derived	thunderstorm electric field E_{amb} and ball lightning height h
Core temperature	T	Derived	Steady-state solution of energy balance equation
Electron density	n_e	Derived	Derived from ionization equilibrium
Bubble radius	R	Derived	Bubble equilibrium and energy density constraint
Lifetime	τ	Derived	$\tau = E_0 / \langle P_{loss} - P_{env} \rangle_t$
Coulomb coupling parameter	Γ	Derived	Electron density, temperature, and particle charge

Parameter reduction path: The original 8 free parameters ($E_0, n_e, T, \eta, Q, C_{turb}, \xi_{env}, \theta$) are compressed to zero through physical derivation — η is replaced by the grounded filament discharge model, Q is derived from conductivity-geometric parameters, n_e is derived from temperature via ionization equilibrium, T is demoted to a steady-state solution determined by E_0 and V_{gap} , C_{turb} is constrained by the standard sphere drag coefficient, $\theta \approx 1$ in the core region, E_0 is derived from lightning energy and ground resistivity, and V_{gap} is derived from thunderstorm electric field and ball lightning height.

4 Formation Mechanism

4.1 Formation Conditions

A critical condition for ball lightning formation is that the ground resistivity struck by lightning is sufficiently high. Define the grounding impedance mismatch:

$$\Xi = \frac{\rho_{ground}}{\eta_{air} \cdot L_{surface}}$$

where ρ_{ground} is the ground resistivity, η_{air} is the near-ground air resistivity, and $L_{surface}$ is the characteristic scale of current diffusion on the ground.

Bifurcation criterion:

- $\Xi \ll 1$ (low-impedance ground) $\Rightarrow$ current flows smoothly into the ground $\Rightarrow$ ordinary lightning;
- $\Xi \gg 1$ (high-impedance ground) $\Rightarrow$ current trapping $\Rightarrow$ possible ball lightning formation.

The rarity of ball lightning corresponds to the low probability of simultaneous high ground resistivity and intense lightning. This also explains why ball lightning frequently occurs in thunderstorm regions over dry, sandy, and rocky ground, and rarely over water or wetlands.

4.2 Formation Chain

Scattering (energy trapping): Lightning strikes high-impedance ground, current cannot flow smoothly into the ground, and part of the return stroke energy is trapped in the near-ground space. Energy trapping efficiency:

$$f_{trap} = \frac{E_0}{E_{lightning}}$$

The typical value is about 10^{-3} – 10^{-2} (0.1%–1% of total lightning energy is trapped), yielding $E_0 \sim 10^6$ – 10^7 J, consistent with independent estimates from the rain barrel event and the Qinghai event.

Vaporization (material injection): The lightning return stroke temperature is approximately 30000 K, far exceeding the temperature required for soil vaporization. Vaporized mass:

$$M_{vapor} \approx \frac{\eta_{heat} \cdot E_{lightning}}{L_{vapor}}$$

A single lightning strike (10^9 J) can vaporize about 1–10 kg of soil, injecting Si, Fe, Ca, and other elements into the plasma plume as nanoparticles.

Charge accumulation and ion linking: Vaporized nanoparticles are repeatedly charged in the intense microwave field, and the surface charge density gradually increases. The Coulomb force between charged particles gradually overcomes the random diffusion of thermal motion, and particles begin to correlate, forming an electromagnetically coupled network. This is a key step from chaos to order.

Energy retention and bubble closure: Intense microwaves propagate in the residual plasma of the streamer channel, and the ponderomotive force repels plasma to form a low-density cavity. When the ponderomotive force balances the plasma thermal pressure, the bubble closes and microwaves are trapped. The energy release rate is determined by the quality factor Q .

4.3 Streamer–Soliton Mapping Function

A mapping relation exists between initial streamer channel parameters and final bubble parameters:

$$R = r_0 \cdot f_R(\Xi, E_{\text{lightning}}), \quad n_e = n_{e0} \cdot f_n(\Xi, \lambda_{mw}), \quad T = T_0 + \Delta T_{\text{microwave}} \cdot f_T(\Xi)$$

Known constraints: expansion ratio $R/r_0 \approx 10^2$ – 10^3 ; mapping functions tend to zero (no bubble) when $\Xi \ll 1$; tend to saturation when $\Xi \gg 1$. The exact functional form awaits calibration by PIC simulations or controlled experiments.

5 Structural Model: Multi-Layer Coupled Plasma Bubble

5.1 Overview of Three-Layer Architecture

Ball lightning is not a uniform plasma sphere, but a precisely coupled three-layer structure.

Table 3: Three-layer architecture of ball lightning

Layer	Radius/Thickness	Temperature	Electron density	Dominant elements	Core Physics
Core	0.05–0.15 m	10000K	10^{11} – 10^{13}cm^{-3}	Si, Fe, Ca, N, O	Ion circulation pump engine
Shell	10 μm –1 mm	3000–5000K	10^{14} – 10^{15}cm^{-3}	Si, Fe, Ca	Coulomb lattice, electromagnetic confinement
Halo	0.1–0.5 m	2000–4000K	10^9 – 10^{11}cm^{-3}	N ₂ , O ₂	Periodic luminosity variation

5.2 Coupling Between Layers

Energy coupling: Core $\rightarrow$ Shell (thermal conduction + bremsstrahlung heating); Shell $\rightarrow$ Core (microwave ohmic heating); Shell $\rightarrow$ Halo (thermal conduction + line radiation escape); Grounded filament discharge $\rightarrow$ Global (pulsed injection).

Material coupling: Core $\rightarrow$ Shell (ions transported outward by electromagnetic and thermophoretic forces); Shell $\rightarrow$ Core (neutral atoms diffused inward by concentration gradient).

Electromagnetic coupling: Shell confines microwaves; grounded filament discharge occurs between the bottom of the shell and the ground.

5.3 Structural Emergence: From Chaos to Order

The three-layer architecture does not exist at the moment of formation, but spontaneously purified through “chemical fractionation” by the ion circulation pump over hundreds of cycles.

Chaotic phase (moment of formation, $\sim 10^{-3}$ – 10^{-2} s): Vaporized material and air plasma are mixed, with inhomogeneous temperature and disordered ionization.

Self-organization purification phase (tens to hundreds of cycles, $\sim 10^{-2}$ – 0.1 s): Ca (Ca I 6.11 eV) and Fe (Fe I 7.87 eV) with lower ionization energies are preferentially ionized and pushed to the shell, followed by Si (Si I 8.15 eV). N and O (> 12 eV) remain in the core and halo due to high ionization energies. Each cycle pulls the spatial distribution of elements toward a fixed point — fluctuations deviating from it are corrected in the next cycle.

Stable phase (after hundreds of cycles): The three-layer architecture emerges and remains stable. This is a self-organization process of dissipative structures — physically identical to pattern formation in the Belousov–Zhabotinsky reaction in chemistry and the emergence of Bénard convection cells in fluid mechanics.

6 Shell Microstructure: Coulomb Lattice

6.1 Formation Conditions and Physical Nature

The shell is a dense plasma compressed by the ponderomotive force, with an extremely high electron density of 10^{14} – 10^{15} cm $^{-3}$ and moderate temperature (3000–5000 K). Under these conditions, the Coulomb potential energy between charged particles is much larger than the kinetic energy of thermal motion, and the plasma enters a strongly coupled state. The parameter measuring coupling strength is:

$$\Gamma = \frac{q^2}{4\pi\epsilon_0 a k_B T}$$

where q is the particle charge and a is the average interparticle distance. When $\Gamma > 175$, charged particles spontaneously arrange into a regular lattice structure — the Coulomb lattice. This is not a chemical crystal; there are no chemical bonds, only a spatially ordered state self-organized by electromagnetic forces. Typical shell parameters easily allow Γ to exceed this critical value.

6.2 Physical Effects of Coulomb Lattice

Structural toughness: The Coulomb lattice can slide and deform as a whole without losing structural integrity. This explains why ball lightning in the Montana video deforms into a hemisphere without breaking after hitting the ground — the lattice undergoes elastic or plastic deformation under external impact, while long-range Coulomb forces maintain the overall ordered structure.

Intra-shell micro-discharges: The Coulomb lattice is a nearly perfect insulator, but electric fields concentrate at lattice defects (dislocations, vacancies, grain boundaries). When the local electric field exceeds the breakdown threshold, micro-discharges are triggered at defects — this is the microscopic physics of the ~ 1 mm moving sparks inside the shell reported by Nikitin. Sparks move along lattice defects rather than randomly in a uniform medium.

Anomalous viscosity and thermal conductivity: Energy transport in the Coulomb lattice relies on collective vibrational modes (phonons) of the lattice, not random collisions in a rarefied gas. This makes the viscosity and thermal conductivity of the shell much higher than those of ordinary plasma at the same temperature, enhancing the shell's confinement efficiency for the high-temperature core.

Optical properties: The long-range ordered Coulomb lattice can produce Bragg scattering for electromagnetic waves of specific wavelengths, providing detectable characteristic signals for future experimental verification.

6.3 Independent Academic Support: Coulomb Crystal Model of the Hessdalen Phenomenon

The Coulomb lattice is not an ad hoc assumption of B-ECDEC to explain ball lightning. In 2010, Paiva and Taft independently proposed a Coulomb crystal cluster model when studying the unexplained luminous spheres repeatedly appearing in the Hessdalen Valley, Norway. They argued that Hessdalen lights consist of macroscopic Coulomb crystal clusters formed in a dusty plasma environment ionized by α particles from radon decay, powered by the geomagnetic field.

The two independent theories are physically identical at the fundamental level — both are Coulomb crystals spontaneously formed by dusty plasmas under strong coupling conditions. The only difference lies in energy sources and confinement methods: Hessdalen lights are powered by α -particle ionization from radon decay and the geomagnetic field, while the ball lightning shell is powered by lightning energy injection and grounded filament discharge. Once formed, the physical behaviors of Coulomb crystals (toughness, defect discharges, phonon transport) are identical, independent of specific energy sources or confinement mechanisms.

The existence of this independent academic tradition provides strong external support for the shell Coulomb lattice in B-ECDEC.

7 Energy Balance: Competition Between Energy Supply and Dissipation

7.1 Radiation Loss

The ball lightning plasma shell is optically thin, and radiation consists of three mechanisms: Bremsstrahlung:

$$P_{brems} = C_{brems} \cdot n_e \cdot \sum_i (Z_i^2 n_i) \cdot T_e^{1/2} \cdot V_{eff}, \quad C_{brems} \approx 1.69 \times 10^{-32} \text{ W} \cdot \text{m}^3 \cdot \text{K}^{-1/2}$$

Line radiation:

$$P_{line} = \sum_k \sum_{i,j} n_k(i) \cdot A_{ij} \cdot h\nu_{ij}$$

Recombination radiation:

$$P_{recomb} = \sum_k n_e n_k \cdot \alpha_{rec}(T_e) \cdot \langle h\nu_{rec} \rangle \cdot V_{eff}$$

7.2 Grounded Filament Discharge and Energy Supply Redundancy

Grounded filament discharge is the main energy source during ball lightning maintenance. A huge potential difference V_{gap} exists between the bottom of ball lightning and the ground, intermittently breaking down air to form discharge filaments. Grounded filament discharge is not a single channel, but a distributed discharge network with multiple parallel filaments. The discharge frequency and current of each filament are determined by the Paschen breakdown condition.

This energy supply redundancy endows ball lightning with structural toughness: if one filament is interrupted by external disturbance (e.g., ball lightning drifts over dry ground), other filaments can still maintain energy supply, and the system will not collapse instantaneously.

7.3 Thermal Stability

Bremsstrahlung grows slowly as $\propto T^{1/2}$, but line radiation and ionization loss increase exponentially in specific temperature ranges. When the temperature exceeds the line radiation and ionization thresholds, P_{loss} rises sharply, suppressing the ionization-heating positive feedback and ensuring a stable equilibrium point for the system in the high-temperature region.

8 Multi-Stability and Luminosity Variation

8.1 Physical Origin of Multi-Stability

In the energy balance equation $P_{loss}(T) = P_{env}$, the nonlinear characteristics of $P_{loss}(T)$ originate from different growth rates of different dissipation channels in different temperature ranges. The system thus has three characteristic temperature intervals:

Table 4: Characteristic steady states

Steady state	Temperature range	Dominant dissipation	Optical power	Observational verification
Low-temperature (dark soliton)	3000–4000K	Bremsstrahlung	mW level	Montana video (300–600 mW)
Medium-temperature (unstable)	4000–7000K	Rapid growth of line radiation	Unstable	—
High-temperature (bright soliton)	8000K	Line radiation + ionization loss	kW–MW level	Qinghai ball lightning (high radiative power density)

In the low-temperature steady state, the core temperature is insufficient to excite abundant line radiation, energy dissipates slowly mainly through bremsstrahlung, and ball lightning is extremely dim but long-lived. In the high-temperature steady state, the core temperature exceeds the ionization threshold, line radiation is intense, and ball lightning is extremely bright but short-lived.

8.2 Transition Dynamics

The driving forces of transitions are changes in V_{gap} and dust concentration n_{dust} . Transition dynamics can be described by a phenomenological equation:

$$\tau \frac{d\Pi}{dt} = \Pi(1 - \Pi) - \delta \cdot \Pi^2 + \xi(t)$$

where τ is the system response time ($\sim 0.1\text{--}1$ s), δ is the coupling strength of ionization-radiation dissipation, and $\xi(t)$ is a random fluctuation term (from the intermittency of grounded filament discharge and turbulent fluctuations of the environmental electric field).

8.3 Periodic Luminosity Variation

Intermittent pulsed injection from grounded filament discharge $\Rightarrow$ periodic recombination of N_2/O_2 in the halo $\Rightarrow$ synchronous periodic oscillations of luminosity and temperature. Yuan's team has independently confirmed synchronous periodic oscillations of temperature and light intensity in the Qinghai ball lightning spectrum.

9 Penetration of Solid Media

Ball lightning exhibits two forms when penetrating glass: (A) perfect penetration with no visible traces; (B) leaving regular circular holes. A 2024 study clearly indicated that glass damage characteristics point to volume heating — energy is absorbed throughout the glass thickness, not just surface heating; most existing theories cannot satisfy volume heating.

The dielectric induced breakdown mechanism in B-ECDEC provides a unified explanation: ball lightning approaches glass $\Rightarrow$ strong polarized electric field (up to $\sim 10^9$ V/m near the surface) instantly converts glass into a lossy medium $\Rightarrow$ microwave/terahertz waves are absorbed and deposit heat throughout the glass thickness $\Rightarrow$ volume heating is achieved. Perfect penetration occurs when channel energy deposition is below the material melting threshold, and permanent holes are left when above the threshold. B-ECDEC is currently the only existing theory that naturally satisfies the volume heating requirement.

10 Extinction Mechanism

The extinction of ball lightning is the natural collapse of the three-layer architecture when two core conditions are no longer met. Three collapse pathways:

V_{gap} attenuation: Weakening thunderstorm electric field $\Rightarrow P_{env}$ decreases $\Rightarrow$ core temperature drops $\Rightarrow$ ionization rate decreases $\Rightarrow$ ion circulation pump efficiency reduces $\Rightarrow$ shell line radiation weakens $\Rightarrow$ ball lightning dims and extinguishes.

Dust depletion: Dust gradually loses or adsorbs on the shell during circulation and no longer returns to the core $\Rightarrow$ line radiation weakens $\Rightarrow$ system enters “dark soliton” mode $\Rightarrow$ energy eventually dissipates completely.

Grounding coupling rupture: Ball lightning drifts over insulators $\Rightarrow \zeta \rightarrow 0 \Rightarrow P_{env} \rightarrow 0 \Rightarrow$ quasi-stationary state collapses instantaneously $\Rightarrow$ explosion (instantaneous release of stored energy) or silent disappearance.

All three extinction modes are consistent with witness reports. Explosions are instantaneous (< 1 s), while gradual decay can last several seconds to tens of seconds.

11 Falsifiable Predictions

B-ECDEC makes the following falsifiable predictions:

- P1: (Spectrum–Microwave Synchronization): Bright ball lightning with characteristic spectral lines should exhibit periodic spectral oscillations synchronized with microwave bursts of specific frequencies. Absence of synchronized microwave signals would falsify the grounded filament discharge as the driver of luminosity variation.
- P2: (Nonmonotonic Lifetime): A critical dust concentration n_{dust}^* exists. Below this value, stronger V_{gap} leads to longer ball lightning lifetime; far above this value, stronger V_{gap} leads to shorter lifetime.
- P3: (EMP/X-Ray Association): Slow penetration of ordinary soda-lime glass by ball lightning should be accompanied by detectable electromagnetic pulses and/or X-ray bursts. Failure to record such signals in multiple confirmed penetration events would falsify the dielectric induced breakdown mechanism. Positive verification of P3 would be decisive for the theory.
- P4: (Dust Injection–Lifetime Shortening): Injecting Si-, Fe-, Ca-containing nanoparticles into stably formed electromagnetic bubbles in the 2026 experimental setup should significantly shorten subsequent bubble lifetime. Unchanged or prolonged lifetime would falsify the mechanism of ionization dissipation accelerating energy loss. Verification of P4 would rule out the pure vaporized silicon hypothesis.
- P5: (Ground Resistivity Statistics): Ground resistivity at ball lightning locations at the moment of lightning strike should be systematically higher than that at random ordinary lightning grounding points. No significant difference would falsify the mechanism that high ground resistivity is a necessary condition for ball lightning formation.
- P6: (Bragg Scattering): As a Coulomb lattice, the ball lightning shell should produce Bragg scattering for electromagnetic waves of specific wavelengths (on the order of lattice constant, millimeter to submillimeter bands). Detection of characteristic scattering peaks in this band would be direct evidence for the existence of the Coulomb lattice.

12 Data Validation

Key observation status:

Table 5: Fourteen independent data validations

No.	Validation dimension	Key observation	Status
1	Laboratory electromagnetic bubble	2026 Xihe experiment confirms electromagnetic solitons	✓
2	Two-component spectrum	Si/Fe/Ca + N ₂ /O ₂ layered luminescence	✓
3	Temperature stratification	O I $\sim$ 8750 K, Si/Fe $\sim$ 4500 K, continuum $\sim$ 2700 K	✓
4	Energy source core	Stronger high-excitation lines in the core	✓
5	Periodic luminosity variation	Synchronous periodic oscillations of temperature and light intensity	✓
6	Volume heating penetration	Requires volume heating	✓
7	Energy density	10^{10} J/m ³	✓
8	Charge quantity	10^{-3} – 10^{-1} C	✓
9	Intra-shell micro-discharges	$\sim$ 1 mm sparks	✓
10	Low-power ball lightning	mW-level dark solitons	✓
11	High-power ball lightning	Bright solitons	✓
12	Vaporization–material injection	Lightning fulgurite forms glass	✓
13	Movement direction	Moves toward power lines/strong electric fields	✓
14	Environmental electric field driving	Driven by thunderstorm electric fields for discharge and movement	✓

The B-ECDEC theory passes all fourteen independent data validations with zero contradictions.

13 Discussion

13.1 Relationship with Other Theories

B-ECDEC is not a negation of other theories, but a superset integration: the electromagnetic soliton model is the limit of B-ECDEC when $n_{dust} \rightarrow 0$ and $V_{gap} \rightarrow 0$; the vaporized silicon hypothesis is the material coupling module of B-ECDEC, providing material sources and chemically assisted exothermic processes for Si/Fe/Ca characteristic spectra; the skyrmion topological model may describe the deeper topological protection mechanism of the bubble, consistent with electromagnetic bubble confinement.

13.2 Cross-Phenomenon Unification

The Coulomb crystal model of the Hessdalen phenomenon and the shell Coulomb lattice of B-ECDEC are physically identical at the fundamental level, sharing the same core structure — Coulomb crystals

spontaneously formed by dusty plasmas under strong coupling conditions. The only difference lies in energy sources and confinement methods. The Coulomb crystal becomes a unified fundamental physics across phenomena, expanding B-ECDEC from a ball lightning theory to a general theoretical framework describing quasi-stationary dusty plasma structures.

13.3 Known Gaps

All known gaps do not affect the self-consistency of the theory at the level of classical plasma physics and electrodynamics. The exact forms of the streamer–soliton mapping function, energy trapping efficiency, and vaporization efficiency are quantitative details awaiting experimental calibration; critical thresholds for multi-stability transitions and interlayer transport rates can be obtained via numerical simulations; ionization–recombination quantum dynamics, dust particle degenerate plasma effects, and self-organized critical state statistical laws are reserved interfaces for deeper physics, while the classical level is sufficient.

13.4 Theoretical Boundaries and Limitations

B-ECDEC is mainly applicable to ball lightning events associated with strong thunderstorm electric fields. For a small number of reported non-thunderstorm ball lightning events, they can be triggered by extreme electric field events such as seismic piezoelectric effects and high-voltage faults within the framework of “sufficient conditions of energy–field strength–potential difference”. The coexistence, splitting, and merging of multiple balls, and disturbances from secondary lightning return strokes have been identified as low-priority remaining issues, not affecting the core self-consistency of the theory.

14 Conclusion

This paper proposes the B-Environmentally Coupled Dissipative Electromagnetic Bubble Theory (B-ECDEC), which redefines ball lightning as a quasi-stationary dissipative electromagnetic bubble jointly determined by the initial stored energy E_0 and the environmentally coupled potential difference V_{gap} . Built on the framework of electromagnetic ponderomotive self-confined bubbles, the theory uses grounded filament discharge as the environmental energy coupling mechanism, an ion circulation pump to catalyze material spectra, a Coulomb lattice to provide structural toughness, and dielectric induced breakdown to explain solid penetration. All parameters are derived from observable physical quantities through bubble equilibrium and quasi-stationary conditions, achieving zero-free-parameter closure. The theory covers the entire life cycle of ball lightning, from formation via scattering–vaporization–energy retention, to emergence of the three-layer architecture through chaos–self-organization purification, to quasi-stationary maintenance–multi-stability coexistence–external disturbance response, and finally to collapse and extinction.

B-ECDEC inherits the experimentally verified foundation of the electromagnetic soliton model, internalizes the material spectral clues of the vaporized silicon hypothesis, and shares fundamental physics with the Coulomb crystal model of the Hessdalen phenomenon, becoming a general theoretical framework for describing quasi-stationary dusty plasma structures. It has passed all fourteen independent data validations, and its core framework has been directly confirmed by laboratory experiments in 2026. It is currently the only self-consistent theoretical framework that simultaneously satisfies three

critical independent pieces of evidence: the 2012 field spectral observations, the 2026 laboratory soliton verification, and the 2024 volume heating penetration test. Six falsifiable predictions provide clear test paths, and the ultimate scientific fate of the theory is entrusted to experimental verification.

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This paper is a theoretical physics preprint. The B-ECDEC theory was completed in May 2026. Experimental tests and academic criticism are welcome.